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Ultrasonic mass fuel flow meter

Abstract

The subject matter of this specification can be embodied in, among other things, a method of sensing that includes activating a first emitter to emit at least one incident wave, transmitting the incident wave along a buffer rod having a first axial end abutted to the first emitter and a second axial end opposite the first axial end, reflecting a first echo of the incident wave by a gap defined along a portion of the buffer rod, detecting the first echo, determining a first amplitude of the first echo, reflecting a second echo of the incident wave by the second axial end, detecting the second echo, determining a second amplitude of the second echo, and determining a reflection coefficient based on the first amplitude and the second amplitude.

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS (1) This application is a divisional of and claims the benefit of priority to U.S. patent application Ser. No. 16/702,152, filed Dec. 3, 2019, the contents of which are incorporated by reference herein.

TECHNICAL FIELD

(1) This instant specification relates to ultrasonic fluid mass flow sensors.

BACKGROUND

(2) Fluid measurement devices are used for the characterization and operation of fluid control systems. As the dynamic bandwidths, flow ranges, accuracies, and reliabilities of flow measurement devices improve, the potential application landscape of such devices broadens. High dynamic bandwidth flow meters can be used as control system feedback sensors for improving steady state and/or transient accuracy in fuel systems. Ultrasonic flow meters (USFM) are a proven industrial technology that can be leveraged for implementation to aircraft turbine systems.

(3) Existing time of flight ultrasonic flow meters are used in the racing and automotive industries, pipeline custody transfer, industrial flow measurement, and many other applications. However, many of these applications encompass steady-state flow conditions, and their respective applications allow for volumetric flow measurement. In other applications, such as aircraft gas turbine engine applications, the fluid environmental conditions of the fuel delivery system imposes significant design challenges.

SUMMARY

(4) In general, this document describes ultrasonic fluid mass flow sensors.

(5) In one aspect, a sensor includes a sensor housing having an interior surface defining a sensor axis and an axial interior sensor housing cavity having a first axial sensor housing portion having a first cross-sectional area perpendicular to the sensor axis, a second axial sensor housing portion arranged adjacent to the first axial sensor housing portion along the sensor axis and having a second cross-sectional area larger than the first cross-sectional area perpendicular to the sensor axis, and a face extending from the interior surface of the first axial housing portion to the interior surface of the second axial housing portion, a buffer rod having a first axial end and a second axial end opposite the first axial end and having a first axial buffer portion arranged within the first housing portion and having the first axial end, a second axial buffer portion arranged within the second housing portion and abutting the face, and having the second axial end, and a third axial buffer portion, extending axially between the first axial buffer portion and the second axial buffer portion, and having a third cross-sectional area, smaller than the first cross-sectional area, perpendicular to the sensor axis, a cavity defined between the interior surface and the third axial buffer portion, and an acoustic transceiver element acoustically mated to the first end.

(6) Various embodiments can include some, all, or none of the following features. The acoustic transceiver element can be configured to emit a vibration having a predetermined wavelength (λ), and the buffer rod has an axial length of about a round multiple of $n/2\lambda$. The third cross-sectional area can be about half as large as the first cross-sectional area. The sensor can also include a tubular fluid conduit having a first end and a second end opposite the first end and defining a conduit axis, arranged such that the conduit axis is substantially aligned with the sensor axis. The sensor can also include another sensor housing having another interior surface defining another sensor axis and another axial interior sensor housing cavity including another first axial sensor housing portion having another first cross-sectional area perpendicular to the other sensor axis, another second axial sensor housing portion arranged adjacent to the other first axial sensor housing portion along the other sensor axis and having another second cross-sectional area larger than the other first cross-sectional area perpendicular to the other sensor axis, and another face extending from the other interior surface of the other first axial housing portion to the other interior surface of the other second housing portion, another buffer rod having another first axial end and another second axial end opposite the other first axial end and including another first axial buffer portion arranged within the other first housing portion and having the other first axial end, another second axial buffer portion arranged within the other second housing portion and abutting the other face, and having the other second axial end, and another third axial buffer portion, extending axially between the other first axial buffer portion and the other second axial buffer portion, and having another third cross-sectional area, smaller than the other first cross-sectional area, perpendicular to the other sensor axis, another cavity defined between the other interior surface and the other third axial buffer portion, and another acoustic transceiver element acoustically mated to the other first end, wherein the other sensor axis is substantially aligned with the conduit axis. The sensor can also include a fluid housing including a fluid housing interior surface defining an axial fluid housing cavity, a first fluid port in fluidic communication with the axial fluid housing cavity, and a second fluid port in fluidic communication with the axial fluid housing cavity, wherein the tubular fluid conduit is in fluidic communication with the second fluid port and extends axially away from the fluid housing along the conduit axis at the first end, and the sensor housing is arranged within the first fluid housing such that the sensor axis is substantially aligned with the conduit axis. The sensor can also include another sensor housing having another interior surface defining another sensor axis and another axial interior sensor housing cavity including another first axial sensor housing portion having another first cross-sectional area perpendicular to the other sensor axis, another second axial sensor housing portion arranged adjacent to the other first axial sensor housing portion along the other sensor axis and having another second cross-sectional area larger than the other first cross-sectional area perpendicular to the other sensor axis, and another face extending from the other

interior surface of the other first axial housing portion to the other interior surface of the other second housing portion, another buffer rod having another first axial end and another second axial end opposite the other first axial end and including another first axial buffer portion arranged within the other first housing portion and having the other first axial end, another second axial buffer portion arranged within the other second housing portion and abutting the other face, and having the other second axial end, and another third axial buffer portion, extending axially between the other first axial buffer portion and the other second axial buffer portion, and having another third cross-sectional area, smaller than the other first cross-sectional area, perpendicular to the other sensor axis, another cavity defined between the other interior surface and the other third axial buffer portion, another acoustic transceiver element acoustically mated to the other first end, and another fluid housing having another fluid housing interior surface defining another axial fluid housing cavity, another first fluid port in fluidic communication with the other axial fluid housing cavity, and another second fluid port in fluidic communication with the other axial fluid housing cavity, wherein the tubular fluid conduit is in fluidic communication with the other second fluid port and extends axially away from the other fluid housing along the conduit axis at the second end, and the other sensor housing is arranged within the other first fluid housing such that the other sensor axis is substantially aligned with the conduit axis. The acoustic transceiver element can include a piezoelectric element. The sensor can also include a matching layer affixed to the second end and having a thickness of about an odd multiple of $\frac{1}{4} \lambda$. The sensor can also include a backing abutted to the acoustic transceiver element axially opposite the first end.

(7) In another aspect, a sensor system includes a fluid housing having a first fluid housing portion defining a first axial fluid housing cavity and having a first fluid port in fluidic communication with the first axial fluid housing cavity, a second fluid housing portion defining a second axial fluid housing cavity and having a second fluid port in fluidic communication with the second axial fluid housing cavity, and a tubular fluid conduit in fluidic communication with the first fluid port at a first end and in fluidic communication with the second fluid port at a second end opposite the first end, and defining a conduit axis, a first acoustic transceiver element arranged within the first axial fluid housing cavity, axially aligned with the conduit axis, and a second acoustic transceiver element arranged within the second axial fluid housing cavity, axially aligned with the conduit axis.

(8) Various embodiments can include some, all, or none of the following features. The sensor system can also include circuitry configured to activate the first acoustic transceiver element to emit a first incident wave, activate the second acoustic transceiver element to emit a second incident wave, detect, by the first acoustic transceiver element, an echo of the first incident wave, determine a fluid acoustic impedance of a fluid in the tubular fluid conduit based on the echo, detect, by the second acoustic transceiver element, at least a first portion of the first incident wave, determine a first time of flight of the portion of the first incident wave, detect, by the first acoustic transceiver element, at least a second portion of the second incident wave, determine a second time of flight of the second portion, determine a mass fluid flow rate based on the determined fluid acoustic impedance, the determined first time of flight, and the determined second time of flight. One or both of the first sensor or the second sensor can include a sensor housing having an interior surface defining a sensor axis and an axial interior sensor housing cavity having a first axial sensor housing portion having a first cross-sectional area perpendicular to the sensor axis, a second axial sensor housing portion arranged adjacent to the first axial sensor housing portion along the sensor axis and having a second cross-sectional area larger than the first cross-sectional area perpendicular to the sensor axis, and a face extending from the interior surface of the first axial housing portion to the interior surface of the second housing portion, a buffer rod having a first axial end and a second axial end opposite the first axial end and having a first axial buffer portion arranged within the first housing portion and having the first axial end, a second axial buffer portion arranged within the second housing portion and abutting the face, and having the second axial end, and a third axial buffer portion, extending axially between the first axial buffer portion and the second axial buffer portion,

and having a third cross-sectional area, smaller than the first cross-sectional area, perpendicular to the sensor axis, a cavity defined between the interior surface and the third axial buffer portion, and an acoustic transceiver element acoustically mated to the first end. The acoustic transceiver element can be configured to emit a vibration having a predetermined wavelength (λ), and the buffer rod can have an axial length of about a round multiple of $n/2\lambda$. The third cross-sectional area can be about half as large as the first cross-sectional area. The acoustic transceiver element can include a piezo element. The sensor system can also include a matching layer affixed to the second end and having a thickness of about an odd multiple of $\frac{1}{4}\lambda$. The sensor system can also include a backing abutted to the acoustic transceiver element axially opposite the first end.

(9) In another aspect, a method of sensing includes activating a first emitter to emit at least one incident wave, transmitting the incident wave along a buffer rod having a first axial end abutted to the first emitter and a second axial end opposite the first axial end, reflecting a first echo of the incident wave by a gap defined along a portion of the buffer rod, detecting the first echo, determining a first amplitude of the first echo, reflecting a second echo of the incident wave by the second axial end, detecting the second echo, determining a second amplitude of the second echo, and determining a reflection coefficient based on the first amplitude and the second amplitude.

(10) Various implementations can include some, all, or none of the following features. The method can also include determining a fluid acoustic impedance of a fluid at the second axial end based on the determined reflection coefficient and a predetermined buffer rod acoustic impedance. The method can also include transmitting, at the second axial end, a portion of the incident wave through the fluid to a sensor arranged a predetermined distance away from and opposite the first emitter, wherein the fluid is within a tubular fluid conduit having a predetermined cross-sectional area, detecting, by the second sensor, the portion of the incident wave, determining, based on the detected portion of the incident wave, a first time of flight of the portion of the incident wave, transmitting, by a second emitter, another incident wave through the fluid to the first sensor, detecting, by the first sensor, the other incident wave, and determining, based on the detected other incident wave, a second time of flight of the other incident wave. The method can also include determining at least one of a velocity of the fluid within the tubular fluid conduit or a speed of sound within the fluid based on the first time of flight, the second time of flight, and the predetermined distance. The method can also include determining a mass fluid flow rate based on the predetermined cross-sectional area, and the determined speed of sound. Reflecting the second echo of the incident wave by the second axial end can also include reflecting the second echo by a $\frac{1}{4}\lambda$ matching layer affixed to the second axial end. One or both of the first emitter and the first sensor can include piezo elements. A piezo element can include the first emitter and the first sensor.

(11) In another aspect, a method of protecting a sensor element includes providing a sensor having a sensor housing having an interior surface defining a sensor axis and an axial interior sensor housing cavity having a first axial sensor housing portion having a first cross-sectional area perpendicular to the sensor axis, a second axial sensor housing portion arranged adjacent to the first axial sensor housing portion along the sensor axis and having a second cross-sectional area larger than the first cross-sectional area perpendicular to the sensor axis, and a face extending from the interior surface of the first axial housing portion to the interior surface of the second housing portion, a buffer rod having a first axial end and a second axial end opposite the first axial end and including a first axial buffer portion arranged within the first housing portion and having the first axial end, a second axial buffer portion arranged within the second housing portion and abutting the face, and having the second axial end, and a third axial buffer portion, extending axially between the first axial buffer portion and the second axial buffer portion, and having a third cross-sectional area, smaller than the first cross-sectional area, perpendicular to the sensor axis, and an acoustic transceiver element acoustically mated to the first end, providing a fluid at second axial end, and blocking, by the buffer rod and the sensor housing, fluid flow from the second end to the acoustic transceiver element.

(12) Various implementations can include some, all, or none of the following features. Fluid flow

from the second end to the acoustic transceiver element can be blocked by the sensor housing and the second axial buffer portion. The method can also include applying fluid pressure against the second axial end to produce an axial force against the buffer rod, transmitting, by the buffer rod, the axial force to the sensor housing, and preventing, by the sensor housing, transmission of the axial force to the acoustic transceiver element. The method can also include transmitting, by the second axial portion, the axial force to the face, wherein the face interferes with axial movement of the buffer rod toward the acoustic transceiver element.

(13) The systems and techniques described here may provide one or more of the following advantages. First, a system can provide improved environmental survivability against wide fluid temperature ranges. Second, the system can provide improved environmental survivability against wide fluid pressure ranges. Third, the system can provide improved environmental survivability against harsh fluids. Fourth, the system can provide integral fluid density sensing. Fifth, the system can be relatively unaffected by fluid flow dynamics (e.g., swirl, vortices, instability). Sixth, the system can be used with update rates of 100 Hz or greater, while maintaining accuracy.

(14) The details of one or more implementations are set forth in the accompanying drawings and the description below. Other features and advantages will be apparent from the description and drawings, and from the claims.

Description

DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a cross-sectional diagram an example of an ultrasonic flow measurement system.
- (2) FIG. 2A is a cross-sectional diagram of an example ultrasonic sensor module of the system of FIG. 1.
- (3) FIG. 2B shows a conceptual example of reflective surface areas in the example ultrasonic sensor module of FIG. 2A.
- (4) FIG. 3 shows a conceptual example of incident wave propagation in the module of FIG. 2A.
- (5) FIG. 4 shows a conceptual example of fluid pressure mitigation in the module of FIG. 2A.
- (6) FIGS. 5A-5C show conceptual examples of incident wave traversal in an ultrasonic flow measurement system.
- (7) FIGS. 6A and 6B are graphs that show example incident waves and echoes in the ultrasonic flow measurement system of FIG. 1.
- (8) FIG. 7 is a flow chart that shows an example of a process for determining a fluid reflection coefficient.
- (9) FIG. 8 is a flow chart that shows an example of a process for determining a mass fluid flow.
- (10) FIG. 9 is a flow chart that shows an example of a process for resisting effects of fluid exposure on the acoustic transducer of the module of FIG. 2A.
- (11) FIG. 10 is a schematic diagram of an example of a generic computer system

DETAILED DESCRIPTION

(12) This document describes ultrasonic fluid mass flow sensor (USFM) systems, and techniques for measuring fluid flow characteristics of fluids. In general, the USFM systems described in this system can be used in fluid environments that would degrade or destroy existing USFMs. The fluid environmental conditions of fuel delivery systems can impose significant design challenges. For current, state of the art, aircraft and other gas turbine engine applications, an ultrasound transducer deployed for such applications will be expected to survive high fluid pressures (e.g., 0 psi to 4000 psi or higher) and a wide range of fluid temperatures, including high fluid temperatures (e.g., -65 degrees F. or lower to 325 degrees F. or higher).

(13) These temperatures and pressures are far more challenging than those required in typical industrial fluid, steam, or pipeline custody transfer applications. To remain effective in such applications, a wetted transducer must also not be degraded by long-term immersion in caustic

fluids such as aircraft fuels and/or additives at high temperatures and/or pressures. The USFM systems described in this document include features that improve the survivability of the USFM under such conditions.

(14) In existing industrial and custody transfer USFMs based on time of flight, cross-correlation, and phase shift measurements have accuracy limitations determined by the flow velocity range, or turn-down ratio, within the flow measurement volume. For example, during low flow conditions the difference between upstream and downstream measurements can be too insensitive to maintain a target accuracy. During high flow conditions, measurement accuracy can suffer from flow instabilities, often caused by the acoustic path being off-axis with respect to the flow, flow separation, and/or non-axisymmetric flow conditions. Off-axis transducer configurations can also cause sensitivity and accuracy problems. Round transducers can impose non-uniform ultrasound fields as the waves pass diagonally through the flow, reducing accuracy. In existing USFMs having ultrasound beams smaller than the flow cross-section, the full flow profile is not insonified and therefore must be estimated, typically with a single K-factor correction value, or a complex coefficient matrix for USFMs using multiple sonic paths, such as in natural gas custody transfer applications. In existing USFM designs, flow measurement accuracy can be difficult to maintain over a large turndown ratio when the flow regime is unstable, or changes substantially from laminar to turbulent flow. For example, some existing industrial USFMs have a practical turndown ratio of no more than 50:1 while maintaining accuracy, even when application piping and flow conditioning are executed ideally. By comparison, a gas turbine fuel system can require a substantially higher turndown of generally 100:1, with some applications upward of 350:1 or more. In addition, a gas turbine flow measurement system must be capable of maintaining dynamic accuracy, with update rates of 100 Hz or more.

(15) Mass flow is critical to the combustion process to maintain a safe and operable fuel to air ratio. Excess fuel to air ratio can lead to compressor surge or over temperature events. Conversely, excess air to fuel can lead to compressor blow out. Either of these events can be detrimental to gas turbine performance and are therefore key design drivers for gas turbine engine design. Additionally, some applications such as gas turbine engines are designed to operate on various fuel types under varying pressures and temperatures.

(16) An important variable, especially in aircraft gas turbine applications, is the variation in fuel specific gravity amidst the fuel types and temperatures. In some applications, the expected fuel specific gravity can vary by approximately 25% across expected temperature ranges and useable fuel types. The wide range in fuel density, if unknown, will drive a broad range in mass fuel flow for a given volumetric flow rate. This variability can lead to large variances in mass air to fuel flow ratios, making engine design across the environmental range inefficient, yielding oversized engines, conservative acceleration and/or deceleration schedules, excessive surge margins, and/or excessive blowout margins.

(17) FIG. 1 is a cross-sectional diagram of an example of an ultrasonic flow measurement (USFM) system **100**. The USFM system **100** includes a fluid housing **110** and two ultrasonic sensor modules **200**. The fluid housing **110** includes an axial fluid housing cavity **120a** defined by an interior surface **121a**, and an axial fluid housing cavity **120b** defined by an interior surface **121b**. A fluid port **122a** defines a fluid path **124a** connected to the fluid cavity **120a**. A fluid port **122b** defines a fluid path **124b** connected to the fluid cavity **120b**. The fluid housing **110** also defines a cavity **126** that extends between the fluid cavity **120a** and the **120b**.

(18) The fluid housing **110** also includes a fluid control conduit **130** that defines a fluid path **132** along a conduit axis **134**. The fluid control conduit **130** fluidically connects the fluid cavity **120a** and the fluid cavity **120b**, putting the fluid cavity **120a** in fluidic communication with the fluid cavity **120b**. The fluid control conduit **130** has a predetermined flowable area **136** and shape (e.g., square, tapered, and/or curved edges, parallel or tapered walls, to affect fluid flow behavior). In some implementations, the fluid housing **110** can be used across many applications, and the fluid

control conduit **130** can be an interchangeable, specialized subcomponent (e.g., an adapter) that can adapt the USFM system **100** for particular fluid types, applications, and/or operational conditions.

(19) Referring now to FIG. 2A, an enlarged cross-sectional diagram of the example ultrasonic sensor module **200** of the system of FIG. 1 is shown. The ultrasonic sensor module **200** includes a sensor housing **202** having an axial interior sensor housing cavity **204** and a sensor axis **206** defined by an interior surface **207**. When the ultrasonic sensor module **200** is assembled to the fluid housing **110** of FIG. 1, the sensor axis **206** is substantially aligned with the conduit axis **134**. The sensor housing **202** has an axial sensor housing portion **208a** having a cross-sectional area **209a** perpendicular to the sensor axis **206**. The sensor housing **202** also has an axial sensor housing portion **208b** having a cross-sectional area **209b** perpendicular to the sensor axis **206**. The cross-sectional area **209b** is dimensionally larger than the cross-sectional area **209a**. A face **210** extends from the interior surface **207** of the axial sensor housing portion **208a** to the interior surface **108** of the axial sensor housing portion **208b**. In the illustrated example, the face **210** is formed as a substantially squared shoulder or ledge at the transition between the cross-sectional area **209a** and the cross-sectional area **209b**. In some embodiments, the face **210** can be a tapered or otherwise non-squared transition between the cross-sectional area **209a** and the cross-sectional area **209b**.

(20) The ultrasonic sensor module **200** also includes an acoustic transceiver element **230**. The acoustic transceiver element **230** is configured to emit acoustic vibrations (e.g., ultrasonic sounds waves) at a predetermined wavelength (λ) when energized. In some embodiments, a separate acoustic driver and acoustic receiver may be implemented as the acoustic transceiver element **230**. In some embodiments, the acoustic transceiver element **230** can be configured to also detect received acoustic vibrations. In some embodiments, the acoustic transceiver element **230** can be a piezo element.

(21) The acoustic transceiver element **230** is acoustically mated with or otherwise abutted to an axial end **252** of a buffer rod **250** by a bonding layer **232**. In some embodiments, the bonding layer **232** can be an adhesive layer. In some embodiments, the buffer rod can be made of any appropriate material or combination of materials that can provide proper acoustic impedance ratios when combined with matching layer material to improve or maximize sensitivity of measurements, are cost effective, can be fabricated within reasonable manufacturing tolerances, and/or provide good mechanical and chemical compatibility in the intended application environment. Examples of buffer rod materials include titanium alloys, austenitic stainless steel, aluminum, borosilicate glasses, fused (e.g., non-crystalline) quartz, and technical ceramics (e.g., AlN, Al.sub.3O.sub.3, SiN, and blends).

(22) In some embodiments, the bonding layer **232** can be omitted, with the acoustic transceiver element **230** in direct contact with the axial end **252**. For example, the acoustic transceiver element **230** can be held in place by a mechanical clamp or other appropriate fixation assembly, or the acoustic transceiver element **230** can be held in place by fixation features formed in the interior surface **207**. In some embodiments, the bonding layer **232** can be formed from a highly ductile material, such as gold or lead, which can be conformed to the mating faces of the acoustic transceiver element **230** and the axial end **252**.

(23) The acoustic transceiver element **230** is backed by a backing **234**. The backing **234** has a predetermined form and is made of a material that improves the sensitivity and/or efficiency of the acoustic transceiver element **230**.

(24) The buffer rod **250** extends along the sensor axis **206** from the axial end **252** to an axial end **254** opposite the axial end **252**. The buffer rod **250** has a predetermined axial length of about a round multiple of one-half of the transmission wavelength of the acoustic transceiver element **230** ($n/2\lambda$). The buffer rod **250** includes an axial buffer portion **256a** arranged within the axial sensor housing portion **208a** and includes the axial end **252**. The buffer rod **250** includes an axial buffer portion **256b** arranged within the axial sensor housing portion **208b** and includes the axial end **254**. In some embodiments, the axial buffer portion **256b** can contact the interior surface directly or

indirectly (e.g., through a seal, sleeve, or bonding material) to substantially seal the sensor cavity **204** from fluid incursion at the axial end **254**.

(25) The buffer rod **250** also includes an axial buffer portion **256c** that extends axially between the axial buffer portion **256a** and the axial buffer portion **256b**. The axial buffer portion **256c** has a cross-sectional area **209c** that is smaller than the cross-sectional area **209a** perpendicular to the sensor axis **206**. A cavity **260** is defined between the interior surface **207** and the axial buffer portion **256c**. The cavity **260** is partly defined by a face **262** defined between the axial buffer portion **256a** and the axial buffer portion **256c**. The face **262** is a predetermined distance from the axial end **252**. Referring to FIG. 2B, the cross-sectional area **209a** is about twice as large as the cross-sectional area **209c**. In other words, the area within axial buffer portion **256c** is about the same as the area of the face **262**.

(26) The buffer rod **250** has a predetermined acoustic impedance ($Z_{\text{sub.buffer}}$). In the illustrated example, the cavity **260** is filled with air (e.g., an air gap), a fluid (e.g., oil), or a solid having an acoustic impedance that is sufficiently different from the acoustic impedance of the buffer rod **250** to reflect an acoustic echo when struck by an acoustic wave (e.g., an ultrasonic ping). In some embodiments, the cavity **260** is evacuated to form at least a partial vacuum.

(27) In the illustrated example, the axial buffer portion **256a** is partly tapered, and is covered by a cladding **270**. The taper has a predetermined shape that is configured to improve the efficiency and/or sensitivity of the ultrasonic sensor module **200** by directing the propagation of incident waves. The cladding **270** is configured to improve the efficiency and/or sensitivity of the ultrasonic sensor module **200** by directing the propagation of incident waves, acoustically isolating the buffer rod **250** from the sensor housing **202**, and/or thermally insulating the buffer rod **250** from the sensor housing **202**. In some embodiments, the taper, the cladding, or both may be omitted. In some embodiments, other portions of the buffer rod **250** may include a cladding.

(28) Referring again to FIG. 2A, the ultrasonic sensor module **200** includes a matching layer **280** acoustically mated with, affixed to, or otherwise abutted to the axial end **252** of the buffer rod **250**. In some embodiments, the matching layer **280** may be adhered to the axial end **252**. In some embodiments, portions of the matching layer **280** may extend to the sensor housing **202** and be affixed (e.g., welded) to the sensor housing **202**. In some embodiments in which the matching layer **280** is affixed to the sensor housing **202**, the joint between the matching layer **280** and the sensor housing **202** can substantially seal the sensor cavity **204** from fluid incursion at the axial end **254**. The matching layer **280** has an axial thickness that is about a round odd multiple of the transmission wavelength of the acoustic transceiver element **230** ($n/4\lambda$), for example $\frac{1}{4}\lambda$.

(29) Referring again to FIG. 1, the two ultrasonic sensor modules **200** face each other across the fluid control conduit **130**. The acoustic transducer elements of the ultrasonic sensor modules **200** are separated by a predetermined distance **150**.

(30) The USFM system **100** includes a controller **190**. The controller **190** includes circuitry configured to activate the ultrasonic sensor modules **200** to cause acoustic incident waves to be emitted, to detect the reception of acoustic waves at the ultrasonic sensor modules **200**, measure the timings between transmission and reception of various combinations of direct and reflected acoustic waves, and/or determine various properties of the USFM system **100** and/or the fluid based in part on those measured timings as will be discussed further in the descriptions of FIGS. 3-9.

(31) In use, a fluid is flowed through the USFM system **100**. For example, a fluid such as fuel can be provided at the fluid port **122a** where it will flow along the fluid path **124a** into the fluid cavity **120a**. The fluid flows around the ultrasonic sensor module **200** to the fluid control conduit **130**. The fluid flows through the fluid control conduit **130** along the fluid path **132** and then flows around the ultrasonic sensor module **200** to the fluid cavity **120b**. The fluid then flows along the fluid path **124b** out the fluid port **122b**. As will be discussed further in the descriptions of FIGS. 3-9, the ultrasonic sensor modules **200** are protected from direct exposure to the fluid, and are used to transmit acoustic waves through the fluid to determine properties of the fluid, such as acoustic

impedance and mass flow.

(32) FIG. 3 shows a conceptual example of incident wave propagation in the ultrasonic sensor module **200** of FIG. 2A. In use, the acoustic transceiver element **230** is activated to emit an incident wave (e.g., a ping). The incident wave is transmitted into and along the buffer rod **250**. A portion of the incident wave, represented by arrow **310**, travels until it encounters the face **262**. The junction of the face **262** and the cavity **260** causes a portion of the incident wave **310** to be reflected as an echo represented by arrow **320**. The echo **320** travels back to be detected by the acoustic transceiver element **230**. In some embodiments, the ultrasonic sensor module **200** may include a separate acoustic emitter and receiver for transmission and detection of the incident waves.

(33) Another portion of the incident wave, represented by arrow **330**, travels until it encounters the axial end **254**. The junction of the axial end and a fluid **301** at the axial end **254** causes a portion of the incident wave **330** to be reflected as an echo represented by arrow **340**. The echo **340** travels back to be detected by the acoustic transceiver element **230**. Another portion of the incident wave, represented by arrow **350**, propagates into the fluid **301** at the axial end **254**.

(34) The time between the transmission of the incident wave and detection of the echo **320** is measured (e.g., by the example controller **190** of FIG. 1) to determine a first time of flight. The time between the transmission of the incident wave and detection of the echo **340** is measured to determine a second time of flight. The amplitudes of the echo **320** and the echo **340** are also measured. As will be discussed further in the descriptions of FIGS. 6A-7, the measured times-of-flight, the measured echo amplitudes, and predetermined information about the acoustic impedance of the buffer rod **250** and predetermined distances between the acoustic transceiver element **230**, the face **262**, and the axial end **254**, can be used to determine properties of the fluid **301** at the axial end **254**, such as acoustic impedance ($Z_{\text{sub.fluid}}$) and/or speed of sound in the fluid ($C_{\text{sub.fluid}}$).

(35) In some implementations, the ultrasonic sensor module **200** can be used in applications other than the USFM system **100**. For example, the ultrasonic sensor module **200** can be put into contact with a fluid (e.g., attached to or submerged in a tank, pipe, or other fluid vessel or volume) and can be sonified as part of a process to determine an acoustic impedance of the fluid, a speed of sound in the fluid, and/or a fluid density of the fluid.

(36) In some implementations, characteristics of the buffer rod **250** itself can be determined based on the measured times-of-flight and/or the measured echo amplitudes (e.g., to calibrate for unknown buffer rod acoustic impedance and/or compensate for the effects of temperature changes on the ultrasonic sensor module **200**). Similarly, in some implementations, the distances between the acoustic transceiver element **230** and one or both of the face **262** and/or the axial end **254** can be determined based on the measured times of flight, the measured echo amplitudes, known distances, known buffer rod acoustic impedance, and/or known buffer rod temperature.

(37) FIG. 4 shows a conceptual example of fluid pressure mitigation in the ultrasonic sensor module **200** of FIG. 2A. In use, the ultrasonic sensor module **200** is at least partly exposed to the fluid **301** at the axial end **254**. In some embodiments, the temperature or chemical properties of the fluid **301** can be damaging to the acoustic transceiver element **230**, therefore the ultrasonic sensor module **200** is configured to prevent the fluid **301** from coming into direct contact with the acoustic transceiver element **230**. For example, direct or indirect (e.g., through a shim, sleeve, cladding, seal, or sealant) contact between the axial buffer portion **256b** and the axial sensor housing portion **208b** and/or between the buffer rod **250** and the face **210** can substantially block fluid flow from the axial end **254** to the acoustic transceiver element **230**. In some implementations, fluid seepage that gets by the buffer rod **250** can be directed to the sensor cavity **204** without contacting a major face of the acoustic transceiver element **230**.

(38) In use, the ultrasonic sensor module **200** is at least partly exposed to fluid pressure, represented by arrows **410**, at the axial end **254**. The fluid pressure **410** is a static fluid pressure relative to the dynamic pressures caused by the acoustic signals used by the acoustic transceiver element **230**. In some embodiments, direct or indirect (e.g., through the buffer rod **250**) application of the fluid

pressure **410** could create a compressive force against the acoustic transceiver element **230** that could offset or otherwise negatively affect signals provided by the acoustic transceiver element **230** in response to sensed acoustic signals. In some implementations, such effects can be compensated for by mathematically by electrically offsetting the sensor signals in order to recover an approximation of the true signal.

(39) The ultrasonic sensor module **200** is configured to prevent the fluid pressure **410** from affecting the acoustic transceiver element **230**. For example, the acoustic transceiver element **230** is acoustically mated to the axial end **252**. As such, the acoustic transceiver element **230** is able to “float” on the buffer rod **250** relative to the sensor housing **202** and not become compressed by the fluid pressure **410**.

(40) The acoustic transceiver element **230** is also protected from the fluid pressure **410** by the mechanical configuration of the buffer rod **250** and the sensor housing **202**. Fluid pressure **410** is applied to the axial end **254**, which urges movement of the buffer rod **250** into the sensor cavity **204**. This pressure that urges such movement is represented by arrows **420**. Movement of the buffer rod **250** is prevented by contact between the axial buffer portion **256b** and the face **210** of the sensor housing **202**, as represented by arrows **430**. As such, the force **420** is prevented from reaching the acoustic transceiver element **230**.

(41) The smaller size of the cross-sectional area **209a** is sized to accommodate acoustic transceiver element **230** and decouple thermal expansion of the sensor housing **202** from the acoustic path. The larger size of the cross-sectional area **209b** is sized to accommodate the pressure induced forces acting on the buffer rod **250**. The transmission of forces into the sensor housing **202** substantially eliminates pressure induced forces from acting on the acoustic transceiver element **230**, substantially eliminating the need for pressure compensation, transducer components that are sized to react pressure induced forces, and/or wetted transducer design constraints.

(42) By decoupling the acoustic transceiver element **230** from the fluid pressure environment, several advantages are observed. For example, fluid/fuel compatibility of the acoustic transceiver element **230** is not required. In another example, the acoustic transceiver element **230** frequency is not restricted by thickness requirements driven by pressure induced forces. In another example in which the acoustic transceiver element **230** is a piezo transducer, the piezo thickness required to support fluid pressure puts the operating frequency of the acoustic transceiver element **230** far below operating requirements of time of flight measurement. In yet another example, the operational frequency of the acoustic transceiver element **230** can be sized to improve acoustic optimization and/or low flow measurement accuracy.

(43) FIGS. 5A-5C show conceptual examples of incident wave traversal in an ultrasonic flow measurement system **500**. In some implementations, the USFM system **500** can be an example of the USFM system **100** of FIG. 1. The USFM system **500** includes two acoustic emitters **510a** and **510b**, two acoustic receivers **512a** and **512b**, and a fluid control conduit **520**. A fluid flows along the fluid control conduit **520** in a direction represented by arrow **501**.

(44) The derivation that follows assumes that the acoustic receivers **512a** and **512b** are aligned with their respective acoustic emitters **510a** and **510b**, perpendicular to the major axis of the fluid control conduit **520**. Therefore, the below derivation omits angles of incidence. If the acoustic emitters **510a**, **510b** and acoustic receivers **512a**, **512b** were placed off axis, the following derivation could be re-derived using an angle of incidence. However, for simplicity, the trigonometry used to compensate for such angles is not used here.

(45) Referring to FIG. 5A, first, the speed of sound traveling through a non-moving fluid is considered:

Distance=Velocity×time

Or:

Length (L)=Speed of sound in fluid (C.sub.fluid)×time(t)

∴L.sub.1=C.sub.fluid×t.sub.1

$$(46) t_1 = \frac{L_1}{C_{\text{fluid}}}$$

(47) Where C_{fuel} is the speed of sound in fluid, $L_{\text{sub.1}}$ is the distance between the acoustic transmitter **510a** and the acoustic receiver **512a**, and $t_{\text{sub.1}}$ is the signal transit time between the acoustic transmitter **510a** and the acoustic receiver **512a**.

(48) Assuming that the direction **501** in which the control volume (fluid) is moving is the same as a direction of sound travel, represented by line **502a** from the acoustic transmitter **510a** to the acoustic receiver **512a**, the speed of the sound wave traveling through the fluid will change relative to the speed of the fluid.

$$\therefore L_{\text{sub.2}} = V_{\text{sub.2}} \times t_{\text{sub.2}}$$

$$V_{\text{sub.2}} = V_{\text{sub.fluid}} + C_{\text{sub.fluid}}$$

$$\therefore L_{\text{sub.2}} = (V_{\text{sub.fluid}} + C_{\text{sub.fluid}}) t_{\text{sub.2}}$$

$$(49) t_2 = \frac{L_2}{(C_{\text{fluid}} + V_{\text{fluid}})}$$

(50) Where $V_{\text{sub.fluid}}$ is the average velocity of moving fluid, $L_{\text{sub.2}}$ is the distance between the acoustic transmitter **510a** and the acoustic receiver **512a**, and $t_{\text{sub.2}}$ is the signal transit time between the acoustic transmitter **510a** and the acoustic receiver **512a**.

(51) Referring now to FIG. 5B, it is assumed that the control volume (fluid) is opposing the direction of the sound travel from the acoustic emitter **510b** to the acoustic receiver **512b**, represented by line **502b**. The speed of the sound wave traveling through the fluid will change relative to the speed of the fluid.

$$\therefore L_{\text{sub.3}} = V_{\text{sub.3}} \times t_{\text{sub.3}}$$

$$V_{\text{sub.3}} = -V_{\text{sub.fluid}} + C_{\text{sub.fluid}}$$

$$\therefore L_{\text{sub.3}} = (-V_{\text{sub.fluid}} + C_{\text{sub.fluid}}) t_{\text{sub.3}}$$

$$(52) t_3 = \frac{L_3}{(C_{\text{fluid}} - V_{\text{fluid}})}$$

(53) Where $L_{\text{sub.3}}$ is the distance between the acoustic emitter **510b** and the acoustic receiver **512b**, and $t_{\text{sub.3}}$ is the signal transit time between the acoustic emitter **510b** and the acoustic receiver **512b**.

(54) Referring for FIG. 5C, for a particular set of ultrasonic sensors, the devices can both emit and receive signals. This means that for a pair of signals, the following characteristics are shared:

(55) $L_{\text{sub.up}} = L_{\text{sub.down}} = L = \text{distance between emitters}$,

(56) $D = \text{diameter} \therefore \text{Area of the fluid control conduit } \mathbf{520}$;

(57) $A = \text{cross section area}$;

(58) $C_{\text{sub.fluid}} = \text{speed of sound in fluid}$;

(59) $V_{\text{sub.fluid}} = \text{velocity of fluid}$;

(60) $\rho_{\text{sub.fluid}} = \text{density of fluid}$;

(61) $Z_{\text{sub.fluid}} = \text{Acoustic impedance of fluid}$.

(62) With the above properties shared, the difference in time between the upstream and downstream signal will allow calculation of various fluid characteristics.

(63) Upstream and downstream transit times become:

$$(64) t_{\text{up}} = \frac{L_{\text{up}}}{(C_{\text{fluid}} - V_{\text{fluid}})} t_{\text{down}} = \frac{L_{\text{down}}}{(C_{\text{fluid}} + V_{\text{fluid}})}$$

(65) Solving for $t_{\text{sub.up}}$, $t_{\text{sub.down}}$, and $C_{\text{sub.fluid}}$:

$$(66) C_{\text{fluid}} = \frac{(L_{\text{down}} - t_{\text{down}} V_{\text{fluid}})}{t_{\text{down}}} C_{\text{fluid}} = \frac{(L_{\text{up}} + t_{\text{up}} V_{\text{fluid}})}{t_{\text{up}}}$$

(67) Since speed of sound is common between the transducers, the speeds of sound are equal to one another and allows fluid velocity to be found:

$$C_{\text{sub.fluid}} = C_{\text{sub.fluid}}$$

$$(68) \frac{L_{\text{down}} - t_{\text{down}} V_{\text{fluid}}}{t_{\text{down}}} = \frac{(L_{\text{up}} + t_{\text{up}} V_{\text{fluid}})}{t_{\text{up}}}$$

$$L_{\text{sub.down}} t_{\text{sub.up}} - t_{\text{sub.down}} t_{\text{sub.up}} V_{\text{sub.fluid}} = L_{\text{sub.up}} t_{\text{sub.down}} + t_{\text{sub.up}} t_{\text{sub.down}} V_{\text{sub.fluid}}$$

$$L_{\text{sub.down}} t_{\text{sub.up}} - L_{\text{sub.up}} t_{\text{sub.down}} = t_{\text{sub.up}} t_{\text{sub.down}} V_{\text{sub.fluid}} + t_{\text{sub.down}} t_{\text{sub.up}} V_{\text{sub.fluid}}$$

$$L_{\text{sub.up}} = L_{\text{sub.down}}$$

$$L(t_{\text{sub.up}} - t_{\text{sub.dn}}) = L_{\text{sup}} - 2V_{\text{sub}} t_{\text{sub.up}} t_{\text{sub.dn}}$$

$$(69) V_{\text{fluid}} = \frac{L(t_{\text{up}} - t_{\text{down}})}{2t_{\text{up}} t_{\text{down}}}$$

(70) Knowing the velocity of the fluid allows the volume fluid flow ($Q_{\text{sub.fluid}}$) to be determined, where $C_{\text{sub.d}}$ is a predetermined discharge coefficient of the fluid in the fluid control conduit **520**:

$$Q_{\text{sub.fluid}} = C_{\text{sub.d}} \times A \times V_{\text{sub.fluid}}$$

(71) Fluid sound speed properties can also be determined. Since the fluid velocity is shared between the pair of transducers the fluid velocity can be solved for. Recalling that:

$$(72) t_{\text{up}} = \frac{L_{\text{up}}}{(C_{\text{fluid}} - V_{\text{fluid}})}$$

And:

$$(73) t_{\text{down}} = \frac{L_{\text{down}}}{(C_{\text{fluid}} + V_{\text{fluid}})}$$

(74) Solving $t_{\text{sub.up}}$ and $t_{\text{sub.dn}}$ for $V_{\text{sub.fluid}}$:

$$V_{\text{sub.fluid}} = (L_{\text{sub.dn}} - t_{\text{sub.dn}} C_{\text{sub.fluid}}) / t_{\text{sub.dn}}$$

$$V_{\text{sub.fluid}} = (-L_{\text{sub.up}} + t_{\text{sub.up}} C_{\text{sub.fluid}}) / t_{\text{sub.up}}$$

(75) Since velocity of the fluid is common between the transducers, the previous two equations equal one another and allow fluid sound speed to be solved:

$$V_{\text{sub.fluid}} = V_{\text{sub.fluid}}$$

$$(76) 0 \frac{(L_{\text{down}} - t_{\text{down}} C_{\text{fluid}})}{t_{\text{down}}} = \frac{(-L_{\text{up}} + t_{\text{up}} C_{\text{fluid}})}{t_{\text{up}}}$$

$$L_{\text{sub.dn}} t_{\text{sub.up}} - t_{\text{sub.dn}} t_{\text{sub.up}} C_{\text{sub.fluid}} = -L_{\text{sub.up}} t_{\text{sub.dn}} + t_{\text{sub.up}} t_{\text{sub.dn}} C_{\text{sub.fluid}}$$

$$L_{\text{sub.dn}} t_{\text{sub.up}} + L_{\text{sub.up}} t_{\text{sub.dn}} = t_{\text{sub.up}} t_{\text{sub.dn}} C_{\text{sub.fluid}} + t_{\text{sub.dn}} t_{\text{sub.up}} C_{\text{sub.fluid}}$$

$$L_{\text{sub.up}} = L_{\text{sub.dn}}$$

$$L(t_{\text{sub.up}} + t_{\text{sub.dn}}) - 2C_{\text{sub.fluid}} t_{\text{sub.up}} t_{\text{sub.dn}}$$

$$(77) C_{\text{fluid}} = \frac{L(t_{\text{up}} + t_{\text{down}})}{2t_{\text{up}} t_{\text{down}}}$$

(78) FIGS. **6A** and **6B** are graphs that show example incident waves and echoes in the ultrasonic flow measurement system of FIG. **1**. FIG. **6A** shows a graph **600** of acoustic amplitude over time, including a sub-duration **601**. FIG. **6B** shows a graph **602** in which the sub-duration **601** has been expanded for visibility.

(79) The graph **600** shows a representation of the emission of an initial incident wave **610** (e.g., when the acoustic transceiver element **230** is activated to send an acoustic “ping”). An echo **620** is received a few milliseconds later. In some implementations, the echo **620** can be the echo **320** of FIG. **3**, which is a reflection of a portion of the incident wave **310** off the face **262** of the cavity **260**.

(80) An echo **630** is received a few milliseconds later. In some implementations, the echo **630** can be the echo **340**, which is a reflection of a portion of the incident wave **330** off the axial end **254**, which is also an interface to the fluid. Echoes **640**, **650**, and **660** represent reverberations in the buffer rod **250**. In operation, the echoes **640-660** can be filtered out or otherwise ignored.

(81) An incident wave **670** represents a portion of the incident wave that is received by an acoustic sensor (e.g., the acoustic transceiver element **230** located downstream or otherwise opposite the acoustic transceiver element **230** that transmitted the incident wave). The amount of time taken by the incident wave **670** to arrive is affected by several variables, such as the fluid density, flow rate, and flow direction of the fluid in the fluid control conduit **130**, and the distance **150**. The amount of time taken for the incident wave **670** can be used as $t_{\text{sub.up}}$ or $t_{\text{sub.dn}}$ (e.g., depending on whether the wave travelled upstream or downstream in the fluid control conduit **130**).

(82) As illustrated in FIG. **4**, the buffer rod **250** is designed to transfer pressure induced forces to the face **210** of the sensor housing **202**. This is achieved through the double diameter construction of the buffer rod **250**, where the smaller cross-sectional area is sized to accommodate the acoustic transceiver element **230** and decouple thermal expansion of the sensor housing **202** from the acoustic path. The larger cross-sectional area of the axial buffer portion **256b** is sized to accommodate the pressure induced forces acting on the buffer rod **250**. The transmission of forces into the sensor housing **202** substantially eliminates pressure induced forces from acting on the

acoustic transceiver element **230** and substantially eliminates the need for (e.g., piezo ceramic) pressure compensation, sizing to react the pressure induced forces, and substantially avoids wetted transducer design constraints.

(83) By decoupling the acoustic transceiver element **230** from the fluid pressure environment, several advantages are observed. For example, fluid/fuel compatibility of the acoustic transceiver element **230** is not required, the acoustic transceiver element **230** frequency is not restricted by thickness requirements driven by pressure induced forces, the thickness of the acoustic transceiver element **230** required to support fluid pressure puts operating frequency far below operating requirements of time of flight measurement, and acoustic transducer frequency can be sized for acoustic optimization and low flow measurement accuracy.

(84) For aircraft turbine fuel systems, mass fuel flow rate can be determined for an understanding of combustion energy content. This is solved through the use of the buffer rod **250**. The internal design of the buffer rod **250** enables additional acoustic benefits which can be intentionally designed into the USFM system **100**. For example, the configuration of the buffer rod **250** enables the controller **190** to determine reflection coefficients for fuel acoustic impedance measurement. This is achieved by introducing a transducer transmit amplitude response (e.g., echoes **320** or **620**), achieved with the cavity **260** which acts as a substantially ideal reflector, and this amplitude can be compared to the return echoes of the buffer rod fluid interface (e.g., echoes **340** or **630**). In some embodiments, the sensitivity of the axial end **254** is further enhanced by the matching layer **280**, however, this will be ignored in order to simplify the equations below.

(85) Fluid acoustic impedance can be determined by setting echo reflection effective areas equal to one another, for example by configuring the cross-sectional areas **209a** and **209c** appropriately. In some implementations, the areas can be non-equal, and a mathematical compensation can be integrated into the process. However, for the sake of clarity, the areas are assumed to be equal in the equations below. This allows for direct measurement of the reflection coefficient. The wave propagation within the buffer rod **250** is articulated such that in air, the echo returned from the face **262** is equivalent to the echo from the axial end **254**.

(86) The reflection coefficient is found through the use of short time Fourier transforms (STFT). The fast Fourier transforms (FFT) of the two echoes are found to determine the peak of the return echoes:

STFT.fwdarw.Amplitude=f(Frequency)

(87) Therefore:

$|A| = |FFT(\text{Echo.sub.1})|_{\text{sub.f}=\text{f.sub.0}}$

$|B| = |FFT(\text{Echo.sub.2})|_{\text{sub.f}=\text{f.sub.0}}$

(88) Where:

(89) Echo.sub.1 is one of the echoes **320** or **620** of FIGS. **3** and **6** respectively, Echo.sub.2 is one of the echoes **340** or **630** of FIGS. **3** and **6** respectively, and f and f.sub.0 are the transducer driving frequency. The reflection coefficient is then found from:

(90) $R = \frac{\text{Math. A} \cdot \text{Math.}}{\text{Math. B} \cdot \text{Math.}}$

(91) And, assuming the buffer rod **250** is in direct interface with the fluid or fuel (e.g., no matching layer **280** in this case):

(92) $R = \frac{Z_2 - Z_1}{Z_2 + Z_1}$

(93) Where R is the reflection coefficient.

$Z_{\text{sub.2}} = Z_{\text{sub.fluid}}$

$Z_{\text{sub.1}} = Z_{\text{sub.buffer}}$

(94) $Z_{\text{fluid}} = \frac{Z_{\text{buffer}}(1 - R)}{1 + R}$

(95) The impedance of the buffer rod **250** can be determined through characterization at the sensor level. With the buffer rod impedance known and the reflection coefficient being measured, the fluid impedance can now be solved for:

$Z_{\text{sub.fluid}} = \rho_{\text{sub.fluid}} C_{\text{sub.fluid}}$

(96) From the equations above, a speed of sound in fluid was solved for. Since fluid impedance and fluid sound speed are known, fluid density can now be solved for.

$$(97) \quad \rho_{\text{fluid}} = \frac{Z_{\text{fluid}}}{C_{\text{fluid}}}$$

Explicitly:

$$(98) \quad \rho_{\text{fluid}} = \frac{\left(\frac{Z_{\text{buffer}}(1-R)}{1+R}\right)}{\left(\frac{L(t_{\text{up}} + t_{\text{down}})}{2t_{\text{up}}t_{\text{down}}}\right)}$$

(99) With volumetric fluid flow and density now known, the mass fluid flow rate can be found:

$$(100) \quad \dot{m}_{\text{fluid}} = Q_{\text{fluid}} \rho_{\text{fluid}} = C_d A V_{\text{fluid}} \rho_{\text{fluid}} \frac{Z_{\text{fluid}}}{C_{\text{fluid}}}$$

(101) FIG. 7 is a flow chart that shows an example of a process 700 for determining a fluid reflection coefficient. In some implementations, the process 700 can be used with the example ultrasonic sensor module 200 of FIGS. 1-2B.

(102) At 710, a first emitter is activated to emit at least one incident wave. For example, the example acoustic transceiver element 230 can be activated to emit an incident wave.

(103) At 720 the incident wave is transmitted along a buffer rod having a first axial end abutted to the first emitter and a second axial end opposite the first axial end. For example, the incident wave can propagate through the buffer rod 250.

(104) At 730 a first echo of the incident wave is reflected by a gap defined along a portion of the buffer rod. For example, the portion of the incident wave 310 can encounter the face 262 of the cavity 260 and be reflected as the echo 320.

(105) At 740, the first echo is detected. For example, the echo 620 of FIGS. 6A and 6B can be detected.

(106) At 750 a first amplitude of the first echo is determined. For example, a FFT can be performed on the echo 620 to determine an amplitude of the echo 620 (e.g., amplitude A as described above).

(107) At 760 a second echo of the incident wave is reflected by the second axial end. For example, the portion of the incident wave 330 is reflected off the axial end 254 as the echo 340. In some implementations, the second echo can be reflected by a $\frac{1}{4} \lambda$ matching layer affixed to the second axial end, for example, the matching layer 280 at the axial end 254.

(108) At 770, the second echo is detected. For example, the echo 630 of FIGS. 6A and 6B can be detected.

(109) At 780, a second amplitude of the second echo is determined. For example, a FFT can be performed on the echo 640 to determine an amplitude of the echo 640 (e.g., amplitude B as described above).

(110) At 790, a reflection coefficient based on the first amplitude and the second amplitude can be determined. For example:

$$(111) \quad R = \frac{\text{Math. A} \cdot \text{Math.}}{\text{Math. B} \cdot \text{Math.}}$$

(112) FIG. 8 is a flow chart that shows an example of a process 800 for determining a mass fluid flow. In some implementations, the process 800 can be used with the example USFM system 100 of FIG. 1.

(113) At 805, a reflection coefficient value is received. For example, the reflection coefficient R determined at 790 can be received.

(114) At 810, a fluid acoustic impedance of a fluid at the second axial end is determined based on the determined reflection coefficient and a predetermined buffer rod acoustic impedance. For example, the reflection coefficient R can be used along with the predetermined buffer rod impedance $Z_{\text{sub.buffer}}$ to determine $Z_{\text{sub.fluid}}$, as described above.

(115) At 815, a portion of the incident wave is transmitted at the second axial end through the fluid to a sensor arranged a predetermined distance away from and opposite the first emitter, where the fluid is within a tubular fluid conduit having a predetermined cross-sectional area. For example, the incident wave 670 of FIG. 6 can travel through the fluid from the ultrasonic sensor module 200 that is upstream to the ultrasonic sensor module 200 that is downstream.

(116) At 820, the second sensor detects the portion of the incident wave. For example, the ultrasonic

sensor module **200** that is downstream can detect the incident wave **670**.

(117) At **825**, a first time of flight of the portion of the incident wave is determined based on the detected portion of the incident wave. For example, $t_{\text{sub.down}}$ can be determined.

(118) At **830**, another incident wave is transmitted, by a second emitter, through the fluid to the first sensor. For example, the ultrasonic sensor module **200** that is downstream can be activated to emit another incident wave upstream.

(119) At **835**, the first sensor detects the other incident wave, and at **840** a second time of flight of the other incident wave is determined based on the detected other incident wave. For example, $t_{\text{sub.up}}$ can be determined.

(120) At **845**, a velocity of the fluid within the tubular fluid conduit is determined. For example, $V_{\text{sub.fluid}}$ can be determined as:

$$(121) V_{\text{fluid}} = \frac{L(t_{\text{up}} - t_{\text{down}})}{2t_{\text{up}}t_{\text{down}}}$$

(122) At **850**, a speed of sound within the fluid is determined. For example, $C_{\text{sub.fluid}}$ can be determined as:

$$(123) C_{\text{fluid}} = \frac{L(t_{\text{up}} + t_{\text{down}})}{2t_{\text{up}}t_{\text{down}}}$$

(124) At **855**, a mass fluid flow rate is determined based on at least the predetermined cross-sectional area, the determined velocity of the fluid, the determined fluid acoustic impedance, and the determined speed of sound. For example:

$$(125) \dot{m}_{\text{fluid}} = Q_{\text{fluid}} = C_d A V_{\text{fluid}} \frac{Z_{\text{fluid}}}{C_{\text{fluid}}}$$

(126) In some implementations, one or both of the first emitter and the first sensor can be piezo elements. In some implementations, the piezo element can include the first emitter and the first sensor. For example, the emitter and sensor can be separate components, or the acoustic transceiver element **230** can perform the emitting and detecting functions within the ultrasonic sensor module **200**.

(127) FIG. **9** is a flow chart that shows an example of a process **900** for resisting effects of fluid exposure on the acoustic transceiver element **230** of the example ultrasonic sensor module **200** of FIGS. **1-4**. At **910**, a sensor is provided. The sensor includes a sensor housing having an interior surface defining a sensor axis and an axial interior sensor housing cavity having a first axial sensor housing portion having a first cross-sectional area perpendicular to the sensor axis, a second axial sensor housing portion arranged adjacent to the first axial sensor housing portion along the sensor axis and having a second cross-sectional area larger than the first cross-sectional area perpendicular to the sensor axis, and a face extending from the interior surface of the first axial housing portion to the interior surface of the second housing portion, a buffer rod having a first axial end and a second axial end opposite the first axial end and having a first axial buffer portion arranged within the first housing portion and having the first axial end, a second axial buffer portion arranged within the second housing portion and abutting the face, and having the second axial end, and a third axial buffer portion, extending axially between the first axial buffer portion and the second axial buffer portion, and having a third cross-sectional area, smaller than the first cross-sectional area, perpendicular to the sensor axis, and an acoustic transceiver element acoustically mated to the first end. For example, the ultrasonic sensor module **200** can be provided.

(128) At **920**, a fluid is provided at the second axial end. For example, the fluid **301**, such as a fuel, can be provided in the fluid cavity **120a** or **120b** so as to contact the axial end **254**.

(129) At **930**, the buffer rod and the sensor housing blocks fluid flow from the second end to the acoustic transceiver element. For example, as discussed in the description of FIG. **4**, the acoustic transceiver element **230** is separated from the fluid **301** by the sensor housing **202** and the buffer rod **250**, and the fluid **301** by the sensor housing **202** and the buffer rod **250** are configured to prevent the fluid **301** from flowing to the acoustic transceiver element **230**.

(130) In some implementations, fluid flow from the second end to the acoustic transceiver element can be blocked by the sensor housing and the second axial buffer portion. For example, the fluid

301 is prevented from flowing to the acoustic transceiver element **230** by interference between the sensor housing **202** and the axial buffer portion **256b**.

(131) At **940**, a fluid pressure is applied against the second axial end to produce an axial force against the buffer rod. For example, the fluid force **410** can be applied against the axial end **254**.

(132) At **950**, the buffer rod transmits the axial force to the sensor housing. For example, the buffer rod **250** transmits the force **420** to the sensor housing **202**.

(133) At **960**, the sensor housing prevents transmission of the axial force to the acoustic transceiver element. In some implementations, the process **900** can also include transmitting, by the second axial portion, the axial force to the face, wherein the face interferes with axial movement of the buffer rod toward the acoustic transceiver element. For example, any movement of the buffer rod **250** into the sensor cavity **204** is prevented by the counteractive force **430** created through contact between the axial buffer portion **256b** and the face **210**.

(134) FIG. **10** is a schematic diagram of an example of a generic computer system **1000**. The system **1000** can be used for the operations described in association with the process **700**, **800**, and/or **900** according to one implementation. For example, the system **1000** may be included in the controller **190**.

(135) The system **1000** includes a processor **1010**, a memory **1020**, a storage device **1030**, and an input/output device **1040**. Each of the components **1010**, **1020**, **1030**, and **1040** are interconnected using a system bus **1050**. The processor **1010** is capable of processing instructions for execution within the system **1000**. In one implementation, the processor **1010** is a single-threaded processor. In another implementation, the processor **1010** is a multi-threaded processor. The processor **1010** is capable of processing instructions stored in the memory **1020** or on the storage device **1030** to display graphical information for a user interface on the input/output device **1040**.

(136) The memory **1020** stores information within the system **1000**. In one implementation, the memory **1020** is a computer-readable medium. In one implementation, the memory **1020** is a volatile memory unit. In another implementation, the memory **1020** is a non-volatile memory unit.

(137) The storage device **1030** is capable of providing mass storage for the system **1000**. In one implementation, the storage device **1030** is a computer-readable medium. In various different implementations, the storage device **1030** may be a floppy disk device, a hard disk device, an optical disk device, or a tape device.

(138) The input/output device **1040** provides input/output operations for the system **1000**. In one implementation, the input/output device **1040** includes a keyboard and/or pointing device. In another implementation, the input/output device **1040** includes a display unit for displaying graphical user interfaces.

(139) The features described can be implemented in digital electronic circuitry, or in computer hardware, firmware, software, or in combinations of them. The apparatus can be implemented in a computer program product tangibly embodied in an information carrier, e.g., in a machine-readable storage device for execution by a programmable processor; and method steps can be performed by a programmable processor executing a program of instructions to perform functions of the described implementations by operating on input data and generating output. The described features can be implemented advantageously in one or more computer programs that are executable on a programmable system including at least one programmable processor coupled to receive data and instructions from, and to transmit data and instructions to, a data storage system, at least one input device, and at least one output device. A computer program is a set of instructions that can be used, directly or indirectly, in a computer to perform a certain activity or bring about a certain result. A computer program can be written in any form of programming language, including compiled or interpreted languages, and it can be deployed in any form, including as a stand-alone program or as a module, component, subroutine, or other unit suitable for use in a computing environment.

(140) Suitable processors for the execution of a program of instructions include, by way of example, both general and special purpose microprocessors, and the sole processor or one of

multiple processors of any kind of computer. Generally, a processor will receive instructions and data from a read-only memory or a random access memory or both. The essential elements of a computer are a processor for executing instructions and one or more memories for storing instructions and data. Generally, a computer will also include, or be operatively coupled to communicate with, one or more mass storage devices for storing data files; such devices include magnetic disks, such as internal hard disks and removable disks; magneto-optical disks; and optical disks. Storage devices suitable for tangibly embodying computer program instructions and data include all forms of non-volatile memory, including by way of example semiconductor memory devices, such as EPROM, EEPROM, and flash memory devices; magnetic disks such as internal hard disks and removable disks; magneto-optical disks; and CD-ROM and DVD-ROM disks. The processor and the memory can be supplemented by, or incorporated in, ASICs (application-specific integrated circuits).

(141) To provide for interaction with a user, the features can be implemented on a computer having a display device such as a CRT (cathode ray tube) or LCD (liquid crystal display) monitor for displaying information to the user and a keyboard and a pointing device such as a mouse or a trackball by which the user can provide input to the computer.

(142) The features can be implemented in a computer system that includes a back-end component, such as a data server, or that includes a middleware component, such as an application server or an Internet server, or that includes a front-end component, such as a client computer having a graphical user interface or an Internet browser, or any combination of them. The components of the system can be connected by any form or medium of digital data communication such as a communication network. Examples of communication networks include, e.g., a LAN, a WAN, and the computers and networks forming the Internet.

(143) The computer system can include clients and servers. A client and server are generally remote from each other and typically interact through a network, such as the described one. The relationship of client and server arises by virtue of computer programs running on the respective computers and having a client-server relationship to each other.

(144) Although a few implementations have been described in detail above, other modifications are possible. In addition, the logic flows depicted in the figures do not require the particular order shown, or sequential order, to achieve desirable results. In addition, other steps may be provided, or steps may be eliminated, from the described flows, and other components may be added to, or removed from, the described systems. Accordingly, other implementations are within the scope of the following claims.

Claims

1. A method of sensing, comprising: activating a first emitter to emit at least one incident wave; transmitting the incident wave along a buffer rod having a first axial end of the buffer rod abutted to the first emitter and a second axial end of the buffer rod opposite the first axial end of the buffer rod; reflecting a first echo of the incident wave by a gap defined along a portion of the buffer rod; detecting the first echo; determining a first amplitude of the first echo; reflecting a second echo of the incident wave by the second axial end of the buffer rod; detecting the second echo; determining a second amplitude of the second echo; and determining a reflection coefficient based on the first amplitude and the second amplitude.
2. The method of claim 1, further comprising determining a fluid acoustic impedance of a fluid at the second axial end of the buffer rod based on the determined reflection coefficient and a predetermined buffer rod acoustic impedance.
3. The method of claim 2, further comprising: transmitting, at the second axial end of the buffer rod, a portion of the incident wave through the fluid to a second receiver arranged a predetermined distance away from and opposite the first emitter, wherein the fluid is within a tubular fluid conduit

having a predetermined cross-sectional area, a first sensor comprises the first emitter and a first receiver, and a second sensor comprises a second emitter and the second receiver; detecting, by the second receiver, the portion of the incident wave; determining, based on the detected portion of the incident wave, a first time of flight of the portion of the incident wave; transmitting, by the second emitter, another incident wave through the fluid to the first receiver; detecting, by the first receiver, the other incident wave; and determining, based on the detected other incident wave, a second time of flight of the other incident wave.

4. The method of claim 3, further comprising determining at least one of a velocity of the fluid within the tubular fluid conduit or a speed of sound within the fluid based on the first time of flight, the second time of flight, and the predetermined distance.

5. The method of claim 4, further comprising determining a mass fluid flow rate based on the predetermined cross-sectional area, and the determined speed of sound.

6. The method of claim 1, wherein reflecting the second echo of the incident wave by the second axial end of the buffer rod further comprises reflecting the second echo by a $\frac{1}{4} \lambda$ matching layer affixed to the second axial end of the buffer rod.

7. The method of claim 1, wherein the first emitter is a piezo element.

8. The method of claim 1, wherein a piezo element comprises the first emitter.

9. The method of claim 1, further comprising: providing a sensor comprising: a sensor housing having an interior surface defining a sensor axis and an axial interior sensor housing cavity comprising: a first axial sensor housing portion having a first cross-sectional area perpendicular to the sensor axis; a second axial sensor housing portion arranged adjacent to the first axial sensor housing portion along the sensor axis and having a second cross-sectional area larger than the first cross-sectional area perpendicular to the sensor axis; and a face extending from the interior surface of the first axial sensor housing portion to the interior surface of the second axial sensor housing portion; wherein the buffer rod further comprises: a first axial buffer portion arranged within the first axial sensor housing portion and comprising the first axial end of the buffer rod; a second axial buffer portion arranged within the second axial sensor housing portion and abutting the face, and comprising the second axial end of the buffer rod; and a third axial buffer portion, extending axially between the first axial buffer portion and the second axial buffer portion, and having a third cross-sectional area, smaller than the first cross-sectional area, perpendicular to the sensor axis; and an acoustic transceiver element acoustically mated to the first axial end of the buffer rod providing a fluid at second axial end of the buffer rod; and blocking, by the buffer rod and the sensor housing, fluid flow from the second axial end of the buffer rod to the acoustic transceiver element.

10. The method of claim 9, wherein fluid flow from the second axial end of the buffer rod to the acoustic transceiver element is blocked by the sensor housing and the second axial buffer portion.

11. The method of claim 9, further comprising: applying fluid pressure against the second axial end of the buffer rod to produce an axial force against the buffer rod; transmitting, by the buffer rod, the axial force to the sensor housing; and preventing, by the sensor housing, transmission of the axial force to the acoustic transceiver element.

12. The method of claim 11, further comprising transmitting, by the second axial buffer portion, the axial force to the face, wherein the face interferes with axial movement of the buffer rod toward the acoustic transceiver element.

13. The method of claim 1, wherein the buffer rod further comprises: a first axial buffer portion comprising the first axial end of the buffer rod having a first cross-sectional area perpendicular to a sensor axis; a second axial buffer portion comprising the second axial end of the buffer rod; and a third axial buffer portion, extending axially between the first axial buffer portion and the second axial buffer portion, and having a third cross-sectional area, smaller than the first cross-sectional area, perpendicular to the sensor axis, wherein the gap is defined between the first axial buffer portion and the second axial buffer portion.

14. The method of claim 13, wherein a sensor comprises: a sensor housing having an interior

surface defining a sensor axis and an axial interior sensor housing cavity comprising: a first axial sensor housing portion having a first cross-sectional housing area perpendicular to the sensor axis, wherein the first axial buffer portion is arranged within the first axial sensor housing portion; a second axial sensor housing portion arranged adjacent to the first axial sensor housing portion along the sensor axis and having a second cross-sectional housing area larger than the first cross-sectional housing area perpendicular to the sensor axis, wherein the second axial buffer portion is arranged within the second axial sensor housing portion; and a face extending from the interior surface of the first axial sensor housing portion to the interior surface of the second axial housing portion, wherein the second axial buffer portion abuts the face; the buffer rod, wherein the gap is defined between the interior surface and the third axial buffer portion; and an acoustic transceiver element acoustically mated to the first axial end of the buffer rod.
